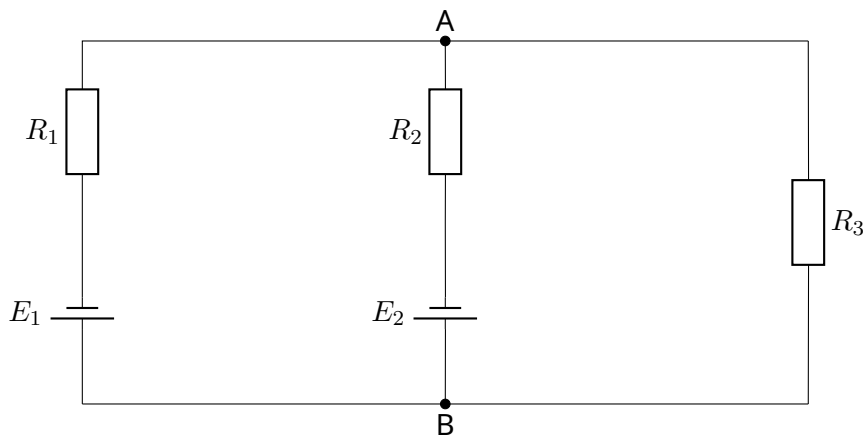


## Exercise Series 1

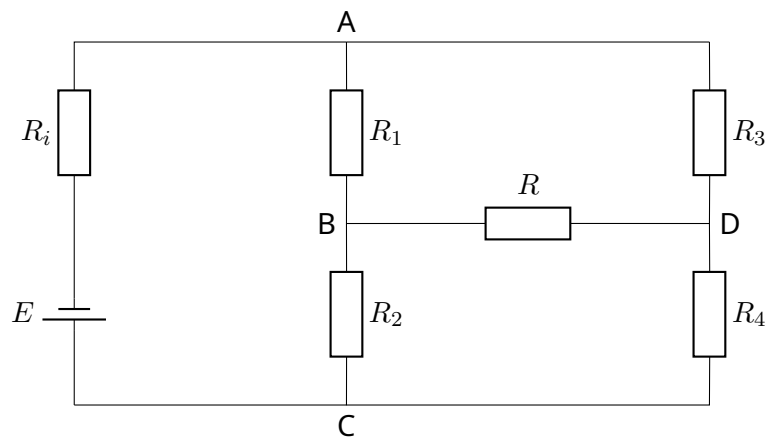
### Exercise 1 (course)

Consider the circuit in the following figure. We aim to determine the current intensities in the three branches. Given that:  $R_1 = 2\ \Omega$ ;  $R_2 = 5\ \Omega$ ;  $R_3 = 10\ \Omega$ ;  $E_1 = 20\text{ V}$ ;  $E_2 = 70\text{ V}$ .



### Exercise 2

Consider the following circuit:



Indicate the number of nodes, the number of branches, and the number of loops.

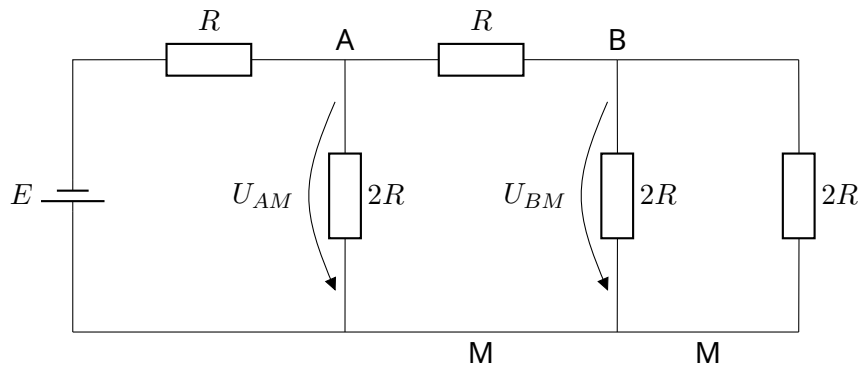
In the following part of the exercise, the BD branch is removed and eliminated from the circuit. We are given  $R_1 = R_2 = R_3 = R_4 = 1\ \Omega$ ,  $R_i = 4\ \Omega$ , and  $E = 10\text{ V}$ . Calculate the currents passing through each resistor.

### Exercise 3 (course)

Using Voltage Divider Theorem:

- Determine  $U_{BM}$  as a function of  $U_{AM}$ .

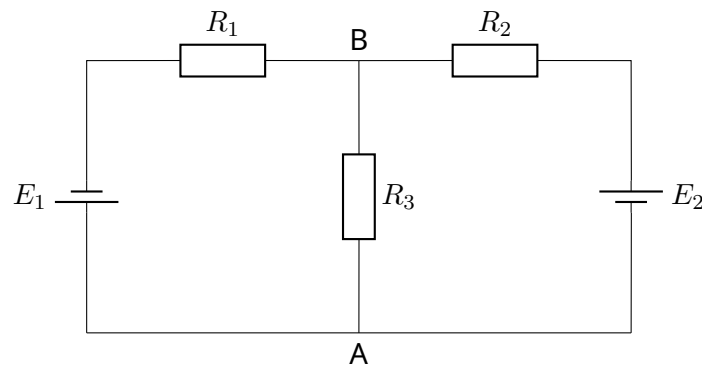
- Determine  $U_{AM}$ , then  $U_{BM}$ , as a function of  $E$ .



#### Exercise 4

Consider the following circuit. We are given:

$$E_1 = 4\text{ V}, E_2 = 24\text{ V}, R_1 = 16\text{ k}\Omega, R_2 = 4\text{ k}\Omega, R_3 = 6\text{ k}\Omega.$$



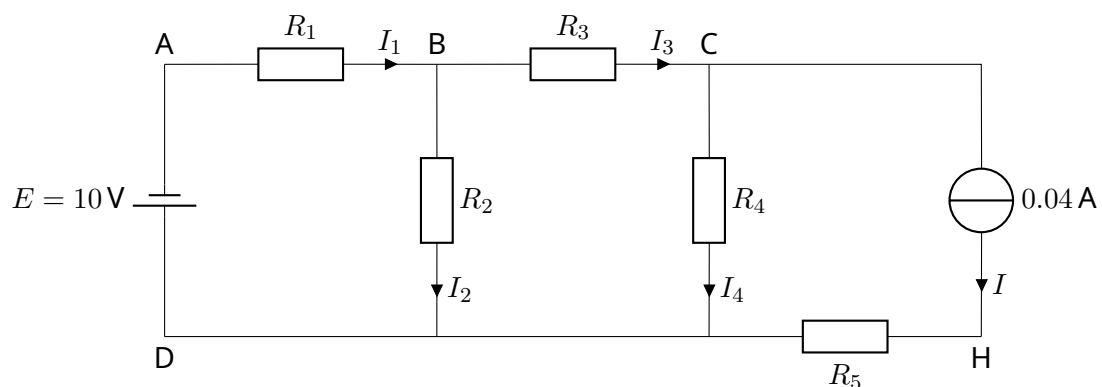
Calculate the current intensity in the AB branch by applying:

- Kirchhoff's Laws
- The Superposition Theorem

#### Exercise 5

Consider the following circuit.

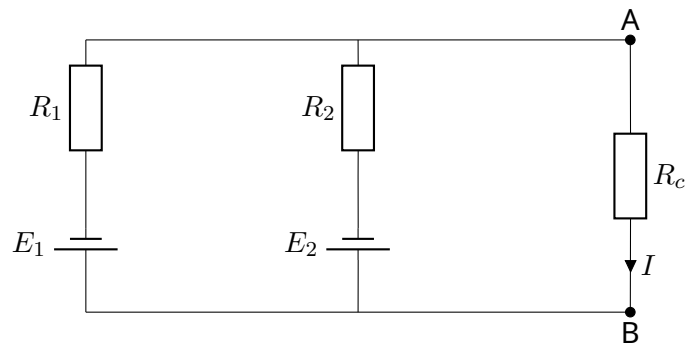
- Using the superposition method, find the currents  $I_1, I_2, I_3$ , and  $I_4$ .
  - Derive the potentials  $V_A, V_B, V_C, V_D$ , and  $V_H$ , taking point D as the origin of the potentials.
- Given:  $R_1 = 100\ \Omega, R_2 = 50\ \Omega, R_3 = 100\ \Omega, R_4 = 50\ \Omega, R_5 = 50\ \Omega$ .



**Exercise 6 (course)**

We consider the electrical circuit given by the following figure:

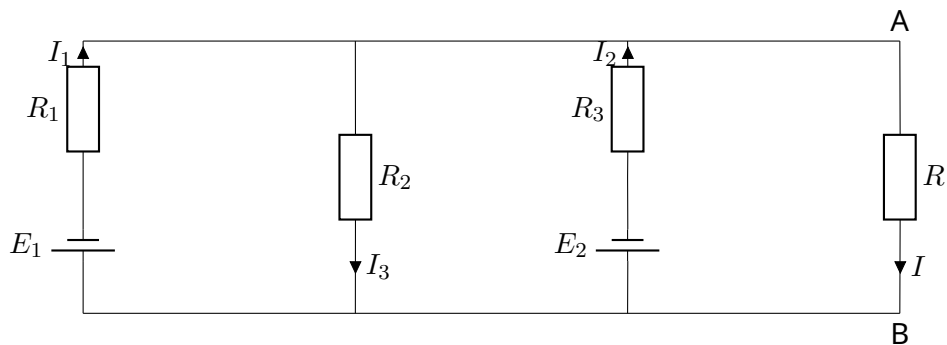
We are given:  $E_1 = 10\text{ V}$ ;  $E_2 = 5\text{ V}$ ;  $R_1 = R_3 = R_4 = 100\ \Omega$ ;  $R_2 = 50\ \Omega$ .



Calculate the current  $I$  by applying Thévenin's theorem.

**Exercise 7**

We consider the given electrical circuit below. The electromotive forces of the generators are given as  $E_1 = 10\text{ V}$  and  $E_2 = 8\text{ V}$ , the resistances  $R_1 = R_2 = R_3 = 200\ \Omega$ , and the load resistance  $R = 100\ \Omega$ .



- Using Norton's theorem, calculate the current that passes through the load  $R$  and deduce the voltage  $U_{AB}$ .
- Using Thévenin's theorem, calculate the voltage  $U_{AB}$  and deduce the current  $I$ .